

Ruoxing (David) Yang

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EDUCATION

Georgetown University

May 2026

B.S. Computer Science, Math (Minor), Economics (Minor) GPA 3.91

Washington, DC

- Experimental AI, Differential Privacy, Deep Reinforcement Learning, LLMs, Approximate Computing

AWS Certified Cloud Practitioner (CLF-C02)

July 2024

WORK EXPERIENCE

Georgetown SecLab (Supervised by Prof. Micah Sherr)

March 2024 (Ongoing)

Research Assistant, Software Developer

Washington, DC

- Developed CADENCE message-passing simulator: implemented MIRAGE and executed simulation experiments, implemented STEALTH logic, redesigned import and driver logics, introduced new datasets.
- Studied and implemented infrastructure for privacy-preserving push-notification system research.

Georgetown University

August 2023 (Ongoing)

Teaching Assistant

Washington, DC

- Held office hours and graded projects for Algorithms, Data Structures, CS 1, CS 2, and Microeconomics.

RESEARCH PUBLICATIONS

“Mirage: Private, Mobility-Based Routing for Censorship Evasion”

March 2024 - August 2025

Network and Distributed System Security Symposium (NDSS) 2026 (Author)

- Helped design MIRAGE routing protocol with Zachary Ratliff, James Mickens, and Micah Sherr. Implemented protocol in CADENCE and handled simulation evaluations.

“DP-Adam-AC”

May 2025 – August 2025

Provost Distinguished Fellow

- Studied and modified small language models (SLMs) and bit-quantized models (bitnet) by applying differentially-private fine-tuning using custom DP-Adam-AC optimization (preprint: [arXiv:2510.05288](https://arxiv.org/abs/2510.05288)).

“A Novel Algorithmic Approach to the Line Addition Problem in Public Transit”

May 2024 – August 2024

Lisa J. Raines Fellow

- Designed an algorithm to improve existing public transit networks (preprint: [arXiv:2412.12109](https://arxiv.org/abs/2412.12109)).

PROJECTS (Code available on GitHub or upon request)

AI/ML Models (Java, Python, Machine Learning)

2024 – 2025

- Implemented Java BP neural network, DQN, propositional logic solver, Bayesian network, PPO, DP-SGD.

Proximal Policy Optimization with Pre-Training (Deep Reinforcement Learning)

2025

- Designed and experimented with PPOPT which modifies PPO for sparse experience RL (preprint: [arXiv:2510.10029](https://arxiv.org/abs/2510.10029))

Video Game Engine (C++, SDL2)

2023

- Created a 2D graphics framework optimized for video-game mechanics.

LEADERSHIP

Provost Distinguished Research Fellow

April 2025 – August 2025

- Led research cohort of 10 fellows and hosted weekly research meetings discussing academic research techniques

Music Director (Nomadic Theatre)

July 2024 – November 2024

- Directed 7 musicians and 13 actors for “Everybody’s Talking About Jamie” musical production.

SKILLS & INTERESTS

- **Programming Languages:** C++/C, Java, Go, Python, SQL, Bash, JavaScript, Typescript, Stata
- **Technologies:** Linux, Latex, Android, IOS, AWS, SDL2, MySQL, GORM, Git, Caddy, Excel, Adobe Indesign
- **Foreign Languages:** English (Native Proficiency), Chinese (Native Proficiency)
- **Interests:** Running, Musical Theatre, Drums, Guitar.